Cancer in Inflammatory Myopathies

Carcinomas are more frequent than sarcomas, and lymphoproliferative malignancies are uncommon. Except for ovarian cancer in women, no specific predominance of cancer types is observed in white individuals with idiopathic inflammatory dermatomyositis (IIDM). In contrast, Asian males most commonly develop nasopharyngeal carcinoma.

Cancer diagnosis is typically based on medical history, review of systems, physical examination, and routine laboratory and imaging studies. There is ongoing debate regarding the use of routine surveillance imaging—such as chest, abdominal, and pelvic CT or MRI—for occult malignancy in this population. Age is a significant factor in malignancy risk. Patients with juvenile-onset classic dermatomyositis (DM) do not appear to have an increased risk of occult cancer. However, individuals aged 50 years or older at the onset of classic DM have a clearly elevated risk of malignancy, estimated at 20% to 30%.

In patients with dermatomyositis and cancer, the leading cause of death is metastatic cancer rather than complications from myositis. Additionally, myositis and cutaneous symptoms in these patients are often less responsive to systemic glucocorticoid therapy. Notably, in some well-documented cases, successful treatment of the underlying malignancy led to resolution of DM activity, while cancer recurrence was associated with reappearance of DM symptoms.

Features that increase the risk of malignancy in adults with classic DM include older age, more extensive and severe skin involvement (e.g., intense generalized erythema ("malignancy suffusion"), blistering, ulceration), and elevated creatine kinase (CK) levels. Conversely, the presence of other systemic manifestations, such as interstitial lung disease, may be negatively correlated with the risk of internal malignancy.

All standard age- and sex-appropriate cancer screening guidelines should be followed in patients with dermatomyositis.

There are currently no population-based data on the risk of internal malignancy in patients with clinically amyopathic dermatomyositis (CADM). A systematic review identified 291 reported cases of adult-onset CADM, of which only 14% were associated with internal malignancy. However, publication bias may influence these findings. One author (R.D.S.) has observed virtually no cases of internal malignancy among approximately 100 patients with adult-onset CADM managed over three decades at three U.S. academic medical centers. Population-based studies are needed to clarify the true malignancy risk in CADM. Until such data are available, it is recommended that patients with CADM undergo the same occult malignancy screening as those with classic DM.

Opportunistic Infections and Lymphoma

Aggressive systemic immunosuppression is often required early in the course of both juvenile- and adult-onset classic DM. Given the chronic or relapsing nature of the disease, patients are frequently exposed to long-term immunosuppressive therapies. This increases the risk of opportunistic infections—including bacterial, mycobacterial, viral, and fungal infections—as well as immunosuppression-related lymphoproliferative disorders, particularly those associated with Epstein-Barr virus.

Effects of Pregnancy

Classic DM and CADM can present for the first time during pregnancy, and pregnancy can complicate pre-existing disease. Onset of DM in the first trimester is generally associated with worse maternal and fetal outcomes compared to onset in the second or third trimesters. The impact of pregnancy on DM is largely influenced by the level of disease activity and the overall health of the mother.

CADM has been reported to develop during the third trimester of pregnancy. Notably, one author (R.D.S.) has observed two sisters with long-standing CADM who each experienced spontaneous remission of skin disease activity during the second and third trimesters of multiple pregnancies, with consistent relapse of skin symptoms within the first three months postpartum, following delivery of healthy infants.